Math 421

Friday, October 31

Announcements

- Homework 7 Due **today**
- Exam 2 – 11/12 5:45-7:15 pm
  - Email me & fill out the form if you have a conflict and need to take the makeup.
  - Sign up for McBurney exam if applicable.

Activity – Find the natural domains

1. $f(x) = \frac{1}{x+5}$

2. $g(x) = \sqrt{2 - x^2}$

3. $\csc x = \frac{1}{\sin x}$

Activity

Recall: $\lim_{x \to a} f(x) = L$ means that the values of $f(x)$ are close to L when the values of x are close to, but not equal to, a.

Sketch, or provide examples of, at least two different ways $\lim_{x \to a} f(x)$ can not exist.

Functional Limits – via $\epsilon - \delta$

Def. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S. Then $\lim_{x \to a} f(x) = L$ if for all $\epsilon > 0$ there exists some $\delta > 0$ such that:

$$\text{if } 0 < |x - a| < \delta \text{ and } x \in S, \text{ then } |f(x) - L| < \epsilon.$$

Activity – Sketch and find δ

1. Let $f(x) = \frac{1}{2}x$, $a = 2$, $L = 1$.
   - If $0 < |x - 2| <$ ______, then $|f(x) - 1| < 1$.
   - If $0 < |x - 2| <$ ______, then $|f(x) - 1| < 1/2$.
2. Let $f(x) = \begin{cases} x + 1, & x \leq 1 \\ 2x, & x > 1 \end{cases}$, $a = 1$, $L = 2$.
   - If $0 < |x - 1| <$ ______, then $|f(x) - 1| < 1$.
   - If $0 < |x - 1| <$ ______, then $|f(x) - 1| < 1/2$.

$\epsilon - \delta$ Proof Recipe

To show that $\lim_{x \to a} f(x) = L$.

1. "Fix $\epsilon > 0$."
2. Determine δ, requires scratch work
3. "Let $\delta = \_\_\_\_\_$"
4. "Assume $0 < |x - a| < \delta$."
5. Show that $|f(x) - L| < \epsilon$.
6. "Since $\epsilon > 0$ was arbitrary, $\lim_{x \to a} f(x) = L$."

Example

Show $\lim_{x \to 2} 6x = 12$.

Discussion.

Proof.

Activity – Fill in the Blanks

Thm. $\lim_{x \to 2} 5x = 10.$

Proof. Fix $\epsilon > 0$. Let $\delta =$ __________. Assume $0 < |x - 2| < \delta$. Then

$|5x - 10| =$ ____

____ $< \epsilon.$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to 2} 5x = 10.$

Example – $\lim_{x \to 1} x^2 + x + 1 = 3$

Discussion.

Algorithm for quadratic polynomials:

1. Factor out $|x - a|$ from $|f(x) - L|$
2. Deal with the extra factor
    a) Write in terms of $x - c$:
    b) Use triangle inequality:
    c) Assume a bound of 1:
3. Let $\delta = \min\{1, \epsilon/\text{answer in c)}\}$

Example – $\lim_{x \to 1} x^2 + x + 1 = 3$

Proof.

Activity – Fill in the Blanks

Thm. $\lim_{x\to 3} x^2 - x = 6.$

Proof. Fix $\epsilon > 0$. Let $\delta = \min\{1, \_\_\_\_\_\_\_\_\}$. Assume $0 < |x-3| < \delta$.

Then $|x^2 - x - 6| = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$

$\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$

$\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ < \epsilon.$

Since $\epsilon > 0$ was arbitrary, $\lim_{x\to 3} x^2 - x = 6.$

Activity – Fill in the Blanks

Thm. If $\lim_{x \to a} f(x) = L$, then $\lim_{x \to a} 4f(x) = 4L$.

Proof. Fix $\epsilon > 0$. Since $\lim_{x \to a} f(x) = L$, there exists $\delta > 0$ such that:

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - L| < \underline{\hspace{3cm}}.$$

Assume $0 < |x - a| < \delta$. Then

$|4f(x) - 4L| =$ ____

____

____ $< \epsilon.$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to a} 4f(x) = 4L$.

Example

Prove using the ϵ-δ definition of the limit that

$$\lim_{x \to 0} x^2 \sin \frac{1}{x} = 0$$

Functional Limits – via Sequences

Def. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S.

Then $\lim_{x \to a} f(x) = L$ if for every sequence (x_n) in S with limit a, we have $\lim_{n \to \infty} f(x_n) = L$.

Example: Let $f(x) = x^2$, $\lim_{x \to 0} x^2 = 0$.

Activity

Prove carefully using sequences that

$$\lim_{x\to 3} x^2 - x = 6.$$

Proof of Equivalence

$(\epsilon - \delta \Rightarrow$ Sequences$)$:

Proof of Equivalence

$(\epsilon - \delta \Leftarrow \text{Sequences})$:

Algebraic Limit Laws

Theorem 20.4. Let f and g be functions such that the limits $L = \lim_{x \to a} f(s)$ and $M = \lim_{x \to a} g(x)$ exist and are finite. Then

i. $\lim_{x \to a}(f + g)(x)$ exists and equals $L + M$.

ii. $\lim_{x \to a}(fg)(x)$ exists and equals LM.

iii. $\lim_{x \to a}(f/g)(x)$ exists and equals L/M provided $M \neq 0$ and $g(x) \neq 0$ for $x \in \text{dom}(g)$.